Parasuram Venkatesh

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Ph.D. student, TIFR-CAM, Bengaluru, India
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Research Interests

Nonlinear PDEs; hyperbolic conservation laws; shock stability; heterogeneous fluxes and nonlocal equations.

Education

Tata Institute of Fundamental Research – Centre for Applicable Mathematics Bengaluru, India
Integrated M.Sc.–Ph.D. in Mathematics 2020–present
Advisor: Shyam Sundar Ghoshal
Master's requirements completed in 2022; Ph.D. expected February 2027

Chennai Mathematical Institute 2016–2019
B.Sc. (Honours) in Mathematics and Computer Science

Research Publications

Published articles

1. *Front Tracking for Scalar Conservation Laws with Spatially Heterogeneous Flux*. Sole author. *Indian Journal of Pure and Applied Mathematics* (2026).
   doi:10.1007/s13226-026-00975-x.
2. *A Non-Conservative, Non-Local Approximation of the Burgers Equation*, with Shyam Sundar Ghoshal and Emil Wiedemann. *Networks and Heterogeneous Media* **20**(4), 1061–1086 (2025).
   doi:10.3934/nhm.2025046.

Preprints

3. *Relative entropies and contraction up to shifts for a nonlocal Burgers equation*, with Nicola De Nitti and Shyam Sundar Ghoshal (2026). CVGMT preprint 8167.
4. *Failure of Uniqueness for Scalar Conservation Laws*, with Shyam Sundar Ghoshal and Abraham Sylla (2026). arXiv:2601.08771.
5. *L^2 Stability of Simple Shocks for Spatially Heterogeneous Conservation Laws*, with Shyam Sundar Ghoshal (2025). arXiv:2502.13687.

Research Software

- **Heterogeneous front tracking.** Python implementation of stationary-profile front tracking for multiplicative heterogeneous fluxes, including sign-changing coefficients. Closed-form event trajectories, coefficient-zero interfaces, diagnostics and explanatory notebooks accompany the analytical construction.
  github.com/parasuramv/heterogeneous-front-tracking
- **Scalar front tracking and random initial data.** Python solver for convex, concave and nonconvex scalar fluxes on the line and circle, with interactive notebooks and integral diagnostics. A companion Streamlit app provides replayable random-data Burgers experiments with boundary-flux mass and entropy diagnostics.
  github.com/parasuramv/scalar-front-tracking Live app: randomized-front-tracking.streamlit.app
- **Möbius conservation-law playground.** Based on my work *Failure of Uniqueness for Scalar Conservation Laws*, this interactive front-tracking app numerically illustrates failure of uniqueness by comparing absorbing and transparent seam rules for the same initial data on a twisted circle. Linked views and diagnostics distinguish finite-level entropy inequalities from weak conservation.
  Live app: mobius-playground.streamlit.app

Teaching Experience

Teaching Assistant, TIFR-CAM Graduate courses

- Advanced Functional Analysis August–December 2026; August–December 2024
- Advanced Partial Differential Equations January–April 2026
- Real Analysis August–December 2023; August–December 2022

Responsibilities included problem-set and examination design, grading, office hours, and individual and small-group problem discussions with cohorts of 8–12 students.

Talks and Selected Meetings

- *Failure of Uniqueness for Scalar Conservation Laws.* Invited talk (forthcoming), Conservation Laws: Theory, Numerics, and Scientific Computing, Indian Institute of Science, Bengaluru, 16–18 December 2026.
- *Failure of Uniqueness for Scalar Conservation Laws.* Talk, TIFR-CAM Alumni Meeting, Bengaluru, 31 October 2026.
- *A Non-Conservative, Non-Local Approximation of the Burgers Equation.* Contributed talk, Mathematical Analysis of Fluid Flows by Variational Methods, Weierstrass Institute, Berlin, 29 September 2025 (meeting: 29 September–1 October).
- Poster presentation, Hyperbolic Problems in Nürnberg, Friedrich-Alexander-Universität Erlangen-Nürnberg, 24–26 September 2025.
- *L^2 Stability of Simple Shocks for Spatially Heterogeneous Conservation Laws.* Opening talk, TIFR-CAM In-house Symposium, Bengaluru, 18 September 2025 (meeting: 18–19 September).
- Participant, Horizons in Nonlinear PDEs, Universität Ulm, 26–30 September 2022.

Outreach

- *Blow-up for Hyperbolic Conservation Laws.* Pizza Talk, TIFR-CAM, 3 December 2025.
- *The Bolzano–Weierstrass Theorem.* SWIM 2025 lecture, Jawaharlal Nehru Planetarium, Bengaluru, 21 July 2025.

Technical Skills and Languages

Python, LaTeX, Lean, Git. Proficient in English and Hindi.

References

- **Shyam Sundar Ghoshal**
  TIFR-CAM, Bengaluru, India.
  `ghoshal@tifrbng.res.in`
- **Nicola De Nitti**
  Department of Mechanics, Mathematics, and Management, Polytechnic University of Bari, Italy.
  `nicola.denitti@poliba.it`
- **Emil Wiedemann**
  Department Mathematik, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany.
  `emil.wiedemann@fau.de`